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From Intention To Implementation: A UTAUT-based Study on AI Adoption In Industry

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Abstract

This study aims to identify the key factors that influence both AI use intention (AIuI) and actual AI use behaviour (AIuB) within industry. By extending the Unified Theory of Acceptance and Use of Technology (UTAUT) with AI-specific determinants. A quantitative survey was conducted among 41 industrial companies in the Netherlands, primarily SMEs. Hypotheses were tested using multiple and stepwise regression analyses, with additional constructs including trust, data management know-how, governance structures, data analytics know-how, and AI experience. AIuB was measured using a four-level AI maturity model. AIuI is significantly influenced by individual perceptions, explaining 49% of the variance. However, AIuB is not predicted by AIuI but instead by AI Experience (AIE) and Data Management (DM) capabilities, which together account for 62% of the variance. Moderation analysis showed no significant effects for gender, age, or SME status. This study contributes to the growing body of work critiquing the predictive power of intention in adoption models. By incorporating AI-specific variables and a maturity-based outcome measure, it offers a more accurate understanding of how organizations move from intention to implementation. Findings show that AI adoption depends not just on intent, but on experiential learning and organizational readiness, calling for a shift from awareness to actionable support in data governance and applied AI skills

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1. Introduction

Artificial Intelligence (AI) is transforming industries worldwide, offering unprecedented opportunities for automation, decision-making, and innovation. A study on AI adoption in businesses in the Netherlands reveals that while AI implementation is growing, it remains inconsistent, with most companies still in the piloting phase and struggling with challenges like transparency, management support, and organizational alignment [1]. Although AI is primarily applied in automation and customer engagement, its full potential has yet to be unlocked, requiring a strategic approach, investment in infrastructure, and stronger collaboration.

However, despite its potential, AI adoption remains inconsistent across sectors, organizations, and regions [2]. While large enterprises increasingly integrate AI into their operations, small and medium-sized enterprises (SMEs) and certain industries still face significant barriers to adoption. SMEs in particular face barriers such as the absence of a clear strategy, limited competencies, weak governance, and a lack of integrated AI solutions. Additionally, high investment costs and the need for well-prepared data present further challenges [3]. Nonetheless, a practical and accessible AI adoption model can help close this adoption gap.

A growing body of literature highlights both enablers and inhibitors of AI adoption. Models such as the Technology-Organization-Environment (TOE) [4] framework and the Unified Theory of Acceptance and Use of Technology (UTAUT) [5] [6] [7] offer theoretical lenses for understanding adoption dynamics. While the TOE framework is often applied to study organizational-level adoption, this paper focuses on UTAUT because it provides constructs that capture individual behavioural intention and use, which are central to our analysis of AI adoption.

The UTAUT model by Venkatesh et al. is highly relevant for investigating technology adoption as it integrates eight existing technology acceptance models. It offers a comprehensive and predictive framework that explains up to 70% of the variance in technology adoption [8]. It identifies four key determinants—performance expectancy, effort expectancy, social influence, and facilitating conditions. Additionally, UTAUT accounts for moderating factors like age, gender, experience, and voluntariness.

One of the main limitations of the UTAUT model is its lack of context-specific factors [9] [10], making it less suited for analysing the unique challenges of AI adoption. While UTAUT effectively predicts general technology acceptance, it does not account for AI-specific concerns. It lacks key determinants essential for AI adoption, such as trust [11], maturity of data management [12] and AI literacy [13] [14]. To improve the model's relevance for AI adoption, it is essential to integrate these domain-specific variables.

The objective of this study is to gain deeper insight into the factors that influence AI adoption within organizations by extending the well-established UTAUT model with AI-specific variables. The study also considers moderating factors like whether the organisation is an SME, gender and age to account for organizational differences in AI maturity and implementation. Through quantitative analysis of survey data, this research seeks to identify the key drivers of AI use intention and behaviour, thereby supporting organizations in navigating the complex process of AI adoption more effectively. The findings will provide actionable insights for organizations, policymakers, and educators to facilitate AI readiness, improve strategic decision-making, and enhance AI-driven innovation.

2. Methodology

2.1. Conceptual Framework

The UTAUT model, identifies four key determinants that influence the acceptance and use of technology [8]:

- Performance Expectancy (PE): the perceived benefits of the technology,
- Effort expectancy (EE): the ease of use,
- Social Influence (SI): the role of peers and organizational pressure, and
- Facilitating Conditions (FC): the availability of support and resources.

In its original form, the UTAUT model effectively explains user intention but is limited in capturing actual usage behaviour—particularly in the context of complex technologies like AI. Therefore, this study distinguishes between

AI use intention (AIuI)—the organization's expressed motivation or plan to adopt AI—and AI use behaviour (AIuB)—the actual implementation and integration of AI technologies within organizational processes.

Recognizing the distinct challenges and requirements of AI adoption, the UTAUT model has been extended with context-specific factors—trust, data management, and data literacy—that have consistently been identified in literature as critical for successful implementation. Trust [11] is essential due to the unclear and complex nature of AI systems.. Trust is therefore conceptualized as a factor directly associated with AI acceptance, rather than merely serving as a facilitating condition. Effective data management [12] ensures that AI systems have access to high-quality, well-structured data. Data literacy [13], particularly knowledge of data analytics, empowers individuals within the organization to understand, interpret, and act upon AI-driven insights Data management, including governance structures and clearly defined roles, functions as a facilitating condition for AI use because it ensures data quality, accountability, and accessibility, which are prerequisites for effective implementation. Similarly, data literacy—such as data analytics expertise and AI experience—plays a comparable role, impacting AI use more directly than the mere intention to adopt it. Accordingly, the general UTAUT model was extended with the following context-specific factors:

- Trust (TR): the perceived reliability and transparency of AI systems,
- Data Management Know-how (DM): company knowledge of data management and governance,
- Data Governance Structure (DGS): the clarity of rules and processes for managing data,
- Roles and Responsibilities (RR): the clarity of how clearly data management tasks and ownership are defined within the organisation,
- Data Analytics Know-how (DA): company knowledge of data analysis techniques,
- AI Experience (AIE): company knowledge of AI applications

In contrast of the original UTAUT model which conceptualizes “experience” as a moderating variable—typically representing general technology familiarity—this study uses AIE as a direct predictor of actual AIuB. This decision is supported by both theoretical reasoning and empirical evidence. Firstly, the construct in this study captures concrete, hands-on organizational experience with AI technologies, rather than general IT proficiency. Secondly, due to the complex, iterative, and skill-intensive nature of AI implementation, experience plays a foundational role in enabling actual use—it acts not merely as a contextual amplifier of intention but is assumed to be a necessary condition for adoption behaviour. Similarly, data management and governance structures were treated as direct predictors, as they provide the foundational infrastructure for AI systems; without them, actual use cannot materialize. This theoretical reasoning is supported by recent studies [15] that highlight organizational data readiness and experiential learning as primary enablers of AI adoption.

In addition to experience, the original UTAUT model incorporates the moderating factors of age, gender, and voluntariness of use, which influence the strength of the relationships between core constructs. In this context, voluntariness of use was excluded, as AI adoption typically occurs at the organizational level and is guided by strategic decisions rather than individual choice, as originally assumed in UTAUT [16]. Unlike traditional technology adoption, AI implementation is primarily driven by leadership, data strategy, and broader organizational priorities. In this study, the AI-specific constructs are defined with an operational focus. While broader organizational factors such as leadership support [17] and change management [18] are acknowledged as closely linked to data governance and management, they were not included as separate constructs. SME status [3] was included as an additional moderating factor, as small and medium-sized enterprises often face distinct resource constraints, governance structures, and decision-making dynamics that may alter the influence of key adoption factors [19]. The modified ATUAT model used in this study is shown in figure 1.

2.2. Hypotheses development

To systematically explore the mechanisms underlying the AIuI and AIuB in industry, a set of hypotheses is developed based on the extended UTAUT framework. The hypotheses are structured around three research questions:

- Q1: Which factors influence the AIuI?
 Q2: Which factors determine the actual AIuB?
 Q3: What is the effect of moderating variables on the relationships with actual AIuB?

Based on individual perceptions, we expect effects of PE (H1), EE (H2), TR (H3), and SI (H4) on AIuI. With respect to factors determining AIuB we test whether AIuI, together with organizational and data-related enablers, predicts AIuB. The enablers include DM (H6), DGS (H7), RR (H8), DA (H9), and AIE (H10), alongside AIuI (H5). Finally, moderation analysis examines whether Gender (H11), Age (H12), or SME status (H13) alter the strength of significant predictor–outcome relationships.

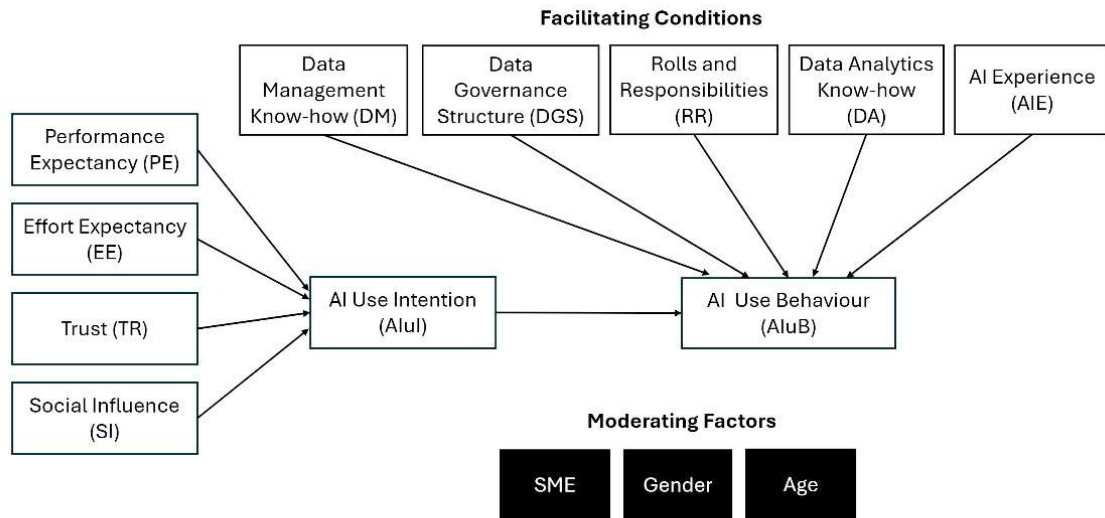


Fig. 1. Modified UTAUT conceptual framework.

2.3. Research Design

A questionnaire (appendix A) was developed to implement the modified UTAUT, focusing on AIuI and AIuB. Except for construct AIuB, which was measured using a four-level AI maturity scale, all other factors were rated their options on a 5-point Likert scale. For measuring organizational AI maturity, we applied the model of Pringle and Zoller [20], which distinguishes between AI Novice, AI Ready, AI Proficient, and AI Advanced. The model has been applied in consultancy and industry contexts as a practical framework for positioning organizations on their AI journey, but it has not yet been systematically validated in academic research; its use here therefore contributes both to its application in an SME context and to exploring its empirical relevance.

Data collection was conducted online from November 2024 to March 2025. A total of 250 companies from the northern Netherlands business network were invited to participate, ensuring representation from organizations engaged in digital and AI-related developments.

To address the research questions and test the corresponding hypotheses, a series of statistical analyses were conducted. Multiple regression analyses were used to examine the influence of key predictors—namely PE, EE, TR, and SI on AIuI, and AIuI, DM, DGS, RR, DA, and AIE on AIuB, in line with research questions 1 and 2. To further investigate the unique contribution of each predictor, two separate stepwise multiple regression analyses were conducted—one for AIuI and one for AIuB. Predictors were entered sequentially based on theoretical relevance and significance in prior models, allowing for the identification of the most influential factors. The models were assessed

by the R-square. The value of R-square < 0.20 , $0.20-0.40$, > 0.40 is considered weak, moderate, and substantial respectively.

The moderation analysis was conducted following the approach described by Warner [21], by including interaction terms between selected predictors and potential moderators. All analyses were performed in SPSS (version 30). Prior to interpretation, multicollinearity was assessed by examining Variance Inflation Factor (VIF) values, which were all well below the threshold of 10, indicating no issues with multicollinearity.

3. Results

Table 1 shows a breakdown of basis characteristics of the respondents including the scores on AIuI and AIuB. The sample consists of 41 respondents, with a majority (88 %) representing (SMEs). Most of the respondent are male (76 %) with 15 % not responding. Regarding age, 59 % are 40 years or older, with also 15 % not responding.

The results of the hypothesis testing, as shown in Table 2, cover both AIuI and AIuB. Hypotheses H1 through H4 examine the influence of individual perceptions—PE, EE, TR, and SI—on AIuI. All four predictors show statistically significant positive effects, with EE (H2) and SI (H4) showing the strongest standardized coefficients ($\beta = 0.51$ and $\beta = 0.37$, respectively). These findings support the hypothesized relationships and highlight the importance of both perceived usability and social context in shaping AIuI.

Table 1. Respondent characteristics

		Frequency	%
SME	Yes	36	87.8
	No	5	12.2
	No response	-	-
	Total	41	100.0
Gender	Male	31	75.6
	Female	4	9.8
	No response	6	14.6
	Total	41	100.0
Age group	39 or younger	11	26.8
	40 or older	24	58.5
	No response	6	14.6
	Total	41	100.0
AI Use Intention (AIuI)	Not at all	0	
	To a small extent	2	4.9
	To some extent	4	9.8
	To a large extent	22	53.7
	Very much	13	31.7
	Total	41	100
AI Maturity (AIuB)	No experience	6	14.6
	Exploratory phase	21	51.2
	Some experience using AI	11	26.8
	Extensive experience and are leading	3	7.3
	Total	41	100

In contrast, hypotheses H5 through H9, which test the effects of AIuI and various organizational and data-related enablers (DM, DGS, RR, DA) on actual AIuB, are not supported. None of these predictors demonstrate a statistically significant effect in the model, suggesting that these factors alone may not directly translate into the AI behavioural level. However, AIE (H10) is found to have a significant positive effect on AIuB ($\beta = 0.51$, $p < 0.01$), indicating that prior familiarity with AI tools plays a key role in actual usage.

The results of the moderation analysis, aimed at testing hypotheses H11a through H13b, show whether gender, age group, and SME status influence the relationships between two key predictors—DM and AIE—and actual AIuB, and are summarized in Table 4. For each hypothesis, an interaction term (e.g., Gender $\times$ DM) was added to the regression model. None of the interaction effects were found to be statistically significant ($p > 0.05$), indicating that the strength or direction of the relationship between the predictors (DM and AIE) and AIuB does not differ meaningfully across gender, age groups, or SME status. In other words, these contextual characteristics do not

appear to moderate the impact of data management knowledge or AI experience on actual AI use. All six hypotheses (H11a–H13b) are therefore not supported. These findings suggest that the effects of data-related knowledge and prior AI experience on AI adoption behaviour are relatively consistent across demographic and organizational subgroups.

The results of two separate stepwise multiple regression analyses—used to identify the most influential predictors of AIuI and actual AIuB—are summarized in Table 3. For AIuI, the model was built incrementally by adding one predictor at each step based on theoretical relevance. TR alone explained 16% of the variance in AIuI ($p = 0.01$). The inclusion EE, SI, and PE in subsequent steps significantly improved the model, resulting in a final explained variance of 49% (adjusted $R^2 = 0.42$). According to the defined thresholds, this indicates a substantial level of explained variance. In other words, nearly half of the AIuI can be explained by four factors: TR, EE, SI, and PE.

For AIuB, the initial model included only AIE, which accounted for 52% of the variance in actual AI use—indicating a substantial and statistically significant effect ($p < 0.001$). The addition of DM in the second model resulted in a statistically significant improvement in explained variance ($\Delta R^2 = 0.10$, $p = 0.01$), contributing to a more substantial model and increasing the total explained variance to 62% (adjusted $R^2 = 0.59$). Meaning, that more than 62% of the AIuB can be explained by only two factors: DM and AIE.

Table 2. Overview of hypothesis testing results for the predictors of AIuI, actual AIuB and interaction effects. The table presents the unstandardized regression coefficients (β) with standard errors, standardized beta values, t-statistics, significance levels, and the resulting support for each hypothesis.

Hypothesis No	Path	Regression coefficient β (standard Error)	Standardized β	t	Sign.	Hypothesis Support
H1	PE→AIuI	0.44 (0.18)	0.38	2.4	0.02*	supported
H2	EE→AIuI	0.47 (0.13)	0.51	3.6	<0.01**	supported
H3	TR→AIuI	0.27(0.12)	0.33	2.3	0.03*	supported
H4	SI→AIuI	0.22 (0.08)	0.37	2.8	<0.01**	supported
H5	AIuI→AIuB	0.01 (0.18)	0.01	0.05	0.96	not supported
H6	DM→AIuB	0.16 (0.17)	0.22	0.94	0.36	not supported
H7	DGS→AIuB	-0.17 (0.13)	-0.23	-1.24	0.23	not supported
H8	RR→AIuB	0.18 (0.15)	0.22	1.13	0.27	not supported
H9	DA→AIuB	0.14 (0.16)	0.20	0.93	0.36	not supported
H10	AIE→AIuB	0.40 (0.14)	0.51	2.83	<0.01**	supported
H11a	Gender x DM→AIuB	0.15 (0.359)	0.19	0.41	0.69	not supported
H11b	Gender x AIE→AIuB	0.22 (0.238)	0.30	0.92	0.37	not supported
H12a	AGE x DM→AIuB	0.38 (0.372)	0.86	1.02	0.31	not supported
H12b	AGE x AIE→AIuB	-075 (0.22)	-0.148	-0.74	0.74	not supported
H13a	SME x DM→AIuB	0.16 (0.24)	0.23	0.70	0.49	not supported
H13b	SME x AIE→AIuB	0.13(0.25)	0.17	0.50	0.62	not supported

* $p < 0.05$, ** $p < 0.001$

These results highlight that while intention-related constructs are central to AIuI, actual AIuB is most strongly associated with prior experience and specific data-related competencies. The inclusion of AIuI in the final model did not result in a statistically significant increase in explained variance, suggesting that AIuI does not contribute meaningfully to the prediction of actual AIuB beyond the effects of AIE and DM.

Table 3. Overview of stepwise regression analyses examining the contribution of individual predictors to AIuI and actual AIuB. For each model step, the table reports explained variance (R^2), adjusted R^2 , the change in R^2 (ΔR^2), F-change, and the corresponding significance level.

Model		Predictors	R ²	Adjusted R ²	ΔR ²	F Change	Sign.F Change
AluI	1	TR	0.16	0.14	0.16	6.61	0.01*
	2	+EE	0.30	0.25	0.13	6.22	0.02*
	3	+SI	0.39	0.33	0.09	4.87	0.03*
	4	+PE	0.49	0.42	0.10	5.83	0.02*
AluB	1	AIE	0.52	0.51	0.52	30.93	<0.001**
	2	+DM	0.62	0.59	0.10	6.94	0.01*

* $p < 0.05$, ** $p < 0.001$

4. Discussion

This study aimed to identify the key factors influencing AIuI and AIuB within organisations, applying a modified UTAUT model tailored to the AI context. The analysis shows that the AIuI is substantially explained by four individual perception factors drawn from the UTAUT model: PE, EE, TR, and SI. Together, these factors account for 49% of the variance in AIuI (adjusted $R^2 = 0.42$), which, based on the applied thresholds, represents a substantial level of explained variance. These results confirm the importance of perception-related determinants in shaping organizational AI adoption intentions. These findings both align with and extend previous research on AI adoption. The significant effects of PE, EE, TR, and SI on AIuI are consistent with earlier studies using the UTAUT model, confirming its continued relevance in understanding technology adoption. As in prior work, the perceived usefulness and ease of use of a technology, along with trust and social pressure, remain critical factors in shaping organizational intention to adopt AI [4] [8].

Actual AIuB appears to be driven by just two key predictors: AIE and DM. Together, these variables account for 62% of the variance in AIuB (adjusted $R^2 = 0.59$), with AIE standing out as the most influential factor. Interestingly, AIuI does not significantly predict actual AI adoption, pointing to a clear disconnect between what organizations aim to do and what they implement. Despite 85% of organizations indicating a strong intention to adopt AI (“to a large extent” or “very much”), real-world implementation remains limited. The assessed AI maturity data show that over half of the organizations are still in the exploratory phase, while only a small proportion (7.3%) classify themselves as advanced users. This mismatch is further corroborated by the regression results, which emphasize that intention alone is not a sufficient driver of adoption—practical experience and data-related capabilities are far more decisive.

The pronounced gap between AIuI and actual AIuB observed in this study reflects a broader challenge in digital transformation: motivation alone is insufficient to drive adoption. From literature this discrepancy can be explained by a combination of practical, organizational, and capability-related barriers [3] [11] [14] [22]. Literature similarly points to this intention–behaviour gap, noting that UTAUT tends to overestimate the direct link between intention and use. Dwivedi et al. argue that intervening factors such as organizational context, culture, and barriers to action weaken this relationship, and call for a revised UTAUT model that explicitly acknowledges the gap. [9] Recent AI adoption studies further confirm that context-specific enablers—particularly related to AI experience (AIE) [13] and data management (DM) [14]—are crucial for actual adoption, underscoring that structural readiness, strategic focus, and experiential learning are required beyond mere intention.

An AI maturity model was adopted to measure actual AIuB more accurately, moving beyond the common limitations of binary (yes/no), duration or frequency intention-based indicators. This approach is supported by Venkatesh et al [22], who advocate for a shift from static to dynamic conceptualizations of technology adoption. They emphasize the importance of examining technology use at the feature level and across different phases—adoption, initial use, and post-adoptive use—to more accurately reflect real-world usage patterns. The applied AI maturity model aligns with this recommendation by framing AI adoption as a developmental process, ranging from no experience to advanced, integrated use. It thus enables a more nuanced understanding of how organizations evolve in their AI capabilities over time, and how contextual factors such as data management, trust, and organizational readiness contribute to this progression. By incorporating this staged approach, the model addresses the intention–behaviour gap identified in prior research and offers a more realistic lens through which to examine the complexities of AI adoption in practice.

This study highlights that prior AIE is the most powerful predictor of actual AIuB, followed by DM. These findings suggest that organizations with practical, hands-on AI experience are significantly more likely to translate intention into real-world implementation. This underscores the critical role of applied learning, data literacy, and structured capacity development in advancing AI adoption. Strengthening organizational competencies in data management—aligned with established frameworks such as DAMA-DMBOK [23]—can lay the groundwork for effective AI integration by improving data quality, establishing clear governance structures, and enabling scalable use of data assets. Training in domains like metadata management, data architecture, and governance not only builds technical capacity but also fosters trust in data-driven processes. For organizations, investing in real-world learning and targeted skill development - such as short training modules and structured readiness assessments - is therefore essential to progress toward AI maturity. For policymakers, these results underline the need to shift from awareness-raising to action-oriented strategies for example by supporting pilot experimentation, enhancing shared data

infrastructure, and embedding maturity frameworks into digital readiness policies to bridge the persistent gap between intention and adoption.

First, the relatively small sample size ($N = 41$) limits statistical power, especially for detecting subtle or interaction effects in the moderation analysis. In addition, several constructs were measured with single-item Likert scales, which may reduce reliability and validity. Future research could strengthen measurement by using validated multi-item scales to capture these constructs more comprehensively. The observed gap between intention and behaviour in AI adoption may partly stem from a level mismatch, with intention typically measured at the individual level and AI maturity reflecting organization-wide capabilities. Addressing this requires frameworks that bridge both perspectives—for example, by combining UTAUT with the TOE model in a multilevel design. Subsequent research could therefore apply multi-level modelling or align intention and behaviour measures at the same analytical level to reduce conceptual misalignment and enrich explanatory power. Lastly, the high barrier to survey participation may itself be indicative of a broader lack of engagement or readiness among organisations when it comes to AI. This introduces the potential for self-selection bias, whereby more digitally mature or AI-aware organisations are disproportionately represented in the sample. As a result, the findings may underrepresent companies that are less advanced or more hesitant in their approach to AI. Given that data governance and management inherently depend on leadership and organizational capacity, future work should further explore how these factors interact to enable or hinder AI adoption.

5. Conclusion

This study explored AI adoption in industry by extending the UTAUT model with AI- and data-specific factors. It demonstrates that successful AI adoption in industry depends less on willingness and more on the ability to act. The results reveal a clear gap between intention and actual use: while AIuI is driven by individual perceptions like EE, PE, TR and SI, actual AIuB is primarily predicted by organizational factors—specifically prior AIE and DM capabilities.

These findings emphasize that moving from intention to execution requires more than positive attitudes. It demands strategic investments in data infrastructure, applied skills, and a culture that supports experimentation. For organizations, this means building the internal conditions necessary for sustained AI integration. For policymakers and educators, the focus must shift toward enabling doing—through practical support, capacity-building programs, and context-aware readiness assessments that translate ambition into concrete outcomes.

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Appendix A. Questionnaire Items

Construct	Question	Scale
Performance Expectancy	To what extent do you expect the use of AI technology to improve your company's performance?	from 1 ("not at all") to 5 ("very much").
Effort expectancy	What are your expectations regarding the effort required to start using AI technology within your company?	from 1 ("very high effort") to 5 ("very low effort").
Trust	How much do you trust AI technology to perform tasks within your company?	from 1 ("no trust") to 5 ("full trust").
Social Influence	To what extent are your company's views on AI technology influenced by external opinions?	from 1 ("not at all") to 5 ("very much").
AI Use Intention	To what extent are you willing to embrace and integrate AI technologies into your daily activities and business processes	from 1 ("not at all") to 5 ("very much").
Data Management Know-how	Does your company have knowledge of data management and associated data governance?	from 1 ("not at all") to 5 ("very much").
Formalised Data Governance Structure	To what extent does your company have a formal data governance structure?	from 1 ("not at all") to 5 ("very much").
Rolls and Responsibilities	To what extent are roles and responsibilities for data management clear within your company?	from 1 ("not at all") to 5 ("very much").
Data Analyses Know-how	Does your company have knowledge of data analysis techniques?	from 1 ("not at all") to 5 ("very much").
AI Experience	Does your company have knowledge of AI technology applications?	from 1 ("not at all") to 5 ("very much").
AI Use Behaviour	How would you rate your company in terms of its maturity in using AI technology?	1. We have no experience with it 2. We are in the exploratory phase of what AI technology could mean for our company 3. We have some experience using AI technology within our company 4. We have extensive experience and are leading in the development, implementation and use of AI technology in our business operations.

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